MILITARY SPECIFICATION

CHALK, MARKING, REFLECTIVE

1. SCOPE

1.1 This specification covers the requirements for a reflex-reflective material in chalk form which can be applied to clothing and later be removed (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-T-111	-	Tape; Paper, Gummed (Sealing and Securing).
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-B-676	-	Boxes, Set-up, Paperboard.

MILITARY

MIL-C-10296	-	Cloth, Cotton, Sateen, Carded.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC 7510

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3. REQUIREMENTS

3.1 Materials.- The reflective chalk shall be made from spherical glass elements which are specially treated to reflect light and are bonded with a suitable organic binding material.

3.1.1 Wrapper.- The wrapper for the chalk shall be suitable for holding the chalk firmly, for protecting the enclosed chalk and shall be easily removed or unwound in small increments from the chalk.

3.2 Dimensions.- The reflective chalk shall be circular in cross section with a diameter of 7/16 inch, minimum, and a length of 1-1/2 inches $\pm$ 1/8 inch.

3.3 Weight.- The weight of a piece of wrapped chalk shall be 10.8 grams minimum, when tested as specified in 4.3.1.

3.4 Reflective intensity.- The reflective intensity of the chalk shall be as specified in table I when tested as specified in 4.3.2.

TABLE I - Reflective intensity at an angle of divergence of 0.2°

Angle of incidence	-4°	0°	10°	20°	30°	40°	50°
	5.0	5.0	5.0	4.5	4.0	3.5	3.0

3.5 Color.- The chalk shall be gray in color.

3.6 Workmanship.- Workmanship shall be such as to meet the requirements of this specification. The quality shall conform to the levels established in this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Inspection shall be in accordance with the provisions in MIL-STD-105 except where otherwise indicated.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards

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unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item.- The end item shall be examined for defects in 4.2.2.1, 4.2.2.2, 4.2.2.3 and 4.2.2.4 at the inspection levels and acceptable quality levels (AQLs) specified in 4.2.2.5. The lot size shall be expressed in units of chalk for the examination in 4.2.2.1 and 4.2.2.2; in units of packages for the examination in 4.2.2.3 and in units of shipping containers for the examination in 4.2.2.4.

4.2.2.1 Visual examination.- The chalk shall be examined for defects in color, appearance and application. The sample unit shall be one piece of chalk. Not more than one sample unit shall be sampled from each package (see 5.1.1).

Examine	Defect
Color	Not gray color Not uniform
Appearance	Shape not cylindrical. Any break, crack or chip (except small chips on periphery of either end). Wrapper missing, torn, not easily unwound in small increments, not continuous or too loosely affixed, causing chalk to fall or slide from wrapper.
Application	Does not leave mark when applied to dark clothing or cloth.

4.2.2.2 Dimensional examination.- The chalk shall be examined for defects in dimension. The sample unit shall be 1 piece of chalk. Not more than one sample unit shall be sampled from each package (see 5.1.1). Samples examined in 4.2.2.1 shall be used for this examination. Any diameter or length that varies from that specified shall be classified a defect.

4.2.2.3 Examination for count of chalk in intermediate packages.- The chalk, packaged for shipment (see 5.1.1) shall be examined to determine compliance with package markings and specified quantity. The sample unit for this examination shall be one intermediate package of chalk. An intermediate package containing less than 20 pieces of chalk shall be classified as a defect.

4.2.2.4 Examination for preparation of delivery.- An examination shall be made to determine that the packaging, packing and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Closed shipping containers shall be examined for closure defects.

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Examine	Defect
Markings (exterior and interior)	Incorrect, incomplete, illegible, omitted, of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged affecting serviceability.
Workmanship	Inadequate application of components, such as; incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling. Bulged or distorted containers.
Content	Number of intermediate packages per shipping container is more or less than required.

4.2.2.5 Inspection levels and acceptable quality levels (AQLs) for examination.- The Inspection levels and the acceptable quality levels (AQLs) expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1	I	6.5
4.2.2.2	I	2.5
4.2.2.3	S-4	2.5
4.2.2.4	S-2	2.5

4.2.3 Testing of the end item.- The end item shall be tested for the characteristics listed in table II. The sample unit shall be 20 pieces of chalk randomly selected throughout the lot. The lot size shall be expressed in units of one package of chalk (20 pieces) each. The sample size shall be as shown below. All test characteristics are applicable to the sample unit. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

Lot size (Units of 20 pieces of chalk)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE II - Instructions for testing

Characteristic	Requirement	Test method	Number of determinations per sample unit	Results reported as numerically to nearest
Weight	3.3	4.3.1	average of 20	0.1 gm
Reflective intensity	3.4	4.3.2	average of 20	0.1

4.3 Tests.-

4.3.1 Weight.- Weigh 20 randomly selected pieces of reflective chalk on a laboratory balance accurate to 0.1 gram. The average weight shall conform to the requirement in 3.3.

4.3.2 Reflective intensity.-

4.3.2.1 Panel preparation.- Tape a 7 by 7 inch square piece of cloth conforming to MIL-C-10296, with the face (filling effect side) exposed to an 8 by 8 inch square clean, smooth metal panel so that the cloth is held taut around all edges and no metal is exposed. Final exposed area of the cloth, after taping, should be 6 by 6 inches square. Flatten the end of the chalk specimen on No. 400 grit sandpaper. Hold the flattened end of the chalk against the cloth and make a series of individual adjoining marks in one direction from the top of the panel to the bottom. When the 6 by 6 inch square area has been filled in with chalk marks, make one mark each across the top and bottom edges at right angles to the previously applied marks. Carefully remove chalk from tape edges. Handle the panels carefully, keep in a horizontal position and do not jar or stack.

4.3.2.2 Apparatus.- A projector, having a maximum lens diameter of 1 inch and capable of projecting a uniform beam of light with an angular spread not more than 1/6 degree shall be used to illuminate a vertical panel. The illumination source shall be a Tungsten lamp of 2850°K minimum color temperature (see 6.5). The light reflected from the panel shall be measured with a photoelectric receiver whose response has been corrected for the color sensitivity of the eye. A selenium barrier layer photocell corrected to the sensitivity of the eye with a Kodak Wratten Filter No. 102 shall be considered proper color correction for the purpose of this test. The maximum diameter of the active area of the receiver shall be 1 inch. The distance from the center of the vertical panel to both the front lens of the projector and the receiver shall be 50 feet plus or minus 2 inches. The arrangement shall be as shown in figure 1.

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4.3.2.3 Test procedure.- Measure the distance from the projector to the panel, the area of the panel (0.25 sq. ft.) and the illumination incident on the panel. Measure the illumination incident on the receiver due to reflection from the vertical panel at the seven angles of incidence (see 6.3) for the 0.2^{o} angle of divergence (see 6.4). The illumination incident on the panel and the receiver shall be measured in the same units and may be measured by the same receiver. Compute the reflective intensity R, from the following equation:

$$R = \frac{L_r D^2}{L_p A}$$

Where R = Reflective intensity.

L_r = Illumination incident upon the receiver.

L_p = Illumination incident upon a plane perpendicular to the incidence ray at the panel position, measured in the same units as L_r.

D = Distance in feet from the panel to the projector.

A = Area in square feet of the panel = 0.25 sq. ft.

The reflective intensity shall be determined on 20 randomly selected pieces of chalk. The average value shall be not less than the minimum requirements in 3.4, table I.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C (see 6.2)

5.1.1 Level A.- Twenty chalks shall be packaged in a snug-fitting folding paperboard box conforming to variety 1, style II, type A, class b or style III, type B, class A or class B of PPP-B-566; or set-up paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676. Closure of full telescope type boxes shall be secured with gummed paper tape conforming to class 2 of UU-T-111 (see 6.3).

5.1.2 Level C.- Chalks shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Level A.- Two thousand chalks, packaged as specified in 5.1 shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V3c or V3s of PPP-B-636. Each shipping container shall be closed, waterproofed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.2 Level B.- Two thousand chalks, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic, variety SW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II in the appendix of the container specification.

5.2.2.1 When specified, shipping containers shall be fabricated from V3c or V3s fiberboard (see 6.2).

5.2.3 Level C.- Chalks, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking.- The following precautionary marking shall appear on one end and side panel of each shipping container in capital letters not less than 3/4-inches in height, as follows:

STORAGE CONDITIONS SHALL NOT
EXCEED 95% RELATIVE HUMIDITY

6. NOTES

6.1 Intended use.- The reflective chalk covered by this specification is intended for temporary markings on clothing of uniformed personnel and other objects which require temporary reflectorization where use of other reflective media are not practical or convenient. The chalk when properly applied provides a brightly visible silver line or area when viewed in a light beam in darkness. Best applications are made on cloth, paper, wood, rubber and other textured or porous materials; however, satisfactory applications have been made on metals, rocks and most non-enamel painted surfaces. Application to very smooth, solid surfaces will result in larger coverage areas; however, brightness values will be less. The markings can be removed from clothing by normal washing or dry cleaning methods and from other surfaces by washing or wiping with a damp cloth.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size and shape (see 3.2).
(c) Selection of applicable levels of packaging and packing. (See 5.1 and 5.2)
(d) Whether V-board shipping containers are required for level B shipments (see 5.2.2.1).

6.3 Incidence angle.- The angle having as its sides the lines normal to the center of the panel and the line joining the center of the panel to the center of the projection lens. If the line normal to the panel is on the same side of the projector as the receiver, the incidence angle is positive. If the line normal to the panel is on the opposite side of the projector from the receiver, the incidence angle is negative.

6.4 Divergence angle.- The angle having as its sides the line joining the center of the panel with the center of the projection lens and the line joining the center of the panel with the center of the receiver.

6.5 A 500 or 1000 watt movie projector has been used as the lightsource.

6.6 The method of packaging referred to in paragraph 5.1.1 is similar to method III of MIL-P-116.

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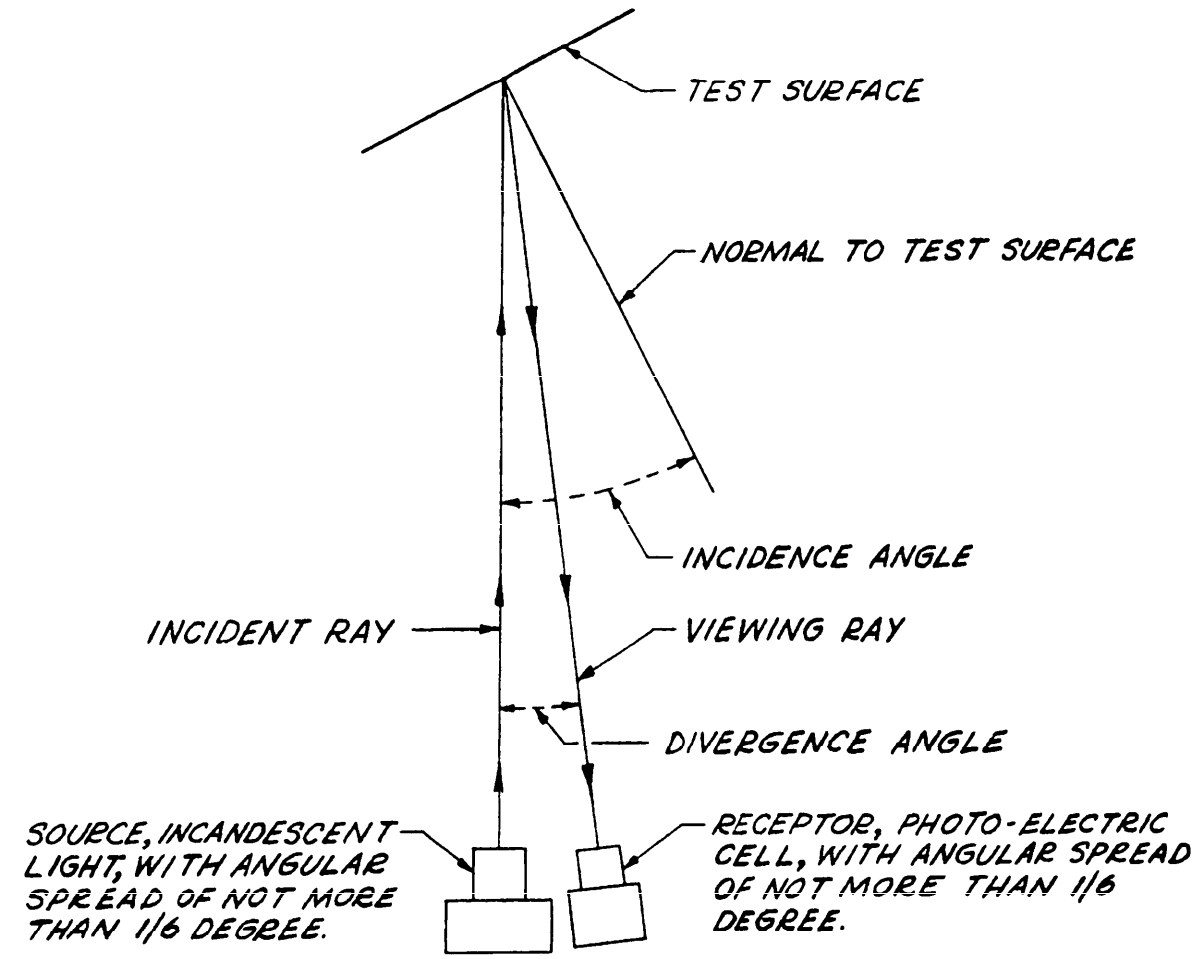
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Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Documents.

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NOTE: ANGLES ARE MEASURED IN THE SAME HORIZONTAL PLANE.

FIGURE 1 - INCIDENCE AND DIVERGENCE ANGLES AS MEASURED IN REFLECTION FACTOR TEST

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: Chalk, Marking, Reflective - MIL-C-43348 (GL)

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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